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(54) **DETERMINING QUANTIZATION SCALE FACTORS FOR LAYERS OF A MACHINE LEARNING MODEL**

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(58) **Field of Classification Search**
CPC G06N 3/086; G06N 3/126; G06N 20/00
See application file for complete search history.

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(57) **ABSTRACT**

Approaches for determining quantization scale factors include generating a population of chromosomes. Each chromosome has multiple genes, and each gene specifies a scale factor associated with a layer of a machine learning model. The population of chromosomes are evaluated, and the evaluating includes, for each chromosome in the population, quantizing floating point weights and floating point values of a representative dataset using the scale factors of the chromosome to produce quantized weights and a quantized dataset in the memory arrangement, initiating processing of the quantized dataset using the quantized weights according to the machine learning model, and gauging a level of accuracy of results produced by the processing of the quantized dataset. Satisfaction of termination criteria is determined based the levels of accuracy associated with the chromosomes in the population. The population of chromosomes is evolved and the evaluating repeated in response to the termination criteria not being satisfied.

20 Claims, 9 Drawing Sheets

